Canvas Template Architecture

Phân tích migration — `rexer-smart-ad-demo` → `rexerai`

Source: https://rexer-smart-ad-demo.vercel.app/studio (80 format, 12 category) **Target:** `src/components/modules/CanvasTemplate/block-editor/` (block + extraElements layer) **Date:** 2026-05-14

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1. Hệ thống hiện tại (snapshot)

Block types (top-level section)

- `hero_split` · `hero_overlay` · `feature_section` · `photo_gallery` · `booking_form` · `cta_section` · `image_carousel`

ExtraElement kinds (free-drag overlay inside 1 block)

- `text` · `edit_text` · `photo` · `video` · `hotspot` · `compare` · `range`

Khả năng compose hiện có: - Drag offset per-element (`elementPositions`) - Motion timeline GSAP per-element - Per-device variants (desktop / mobile) - AI apply-product, AI block-content, AI ImportBlock từ ảnh/Figma

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2. Mapping 80 formats → trạng thái rexerai

2.1. ✅ ĐÃ CÓ (no work needed)

Smart-ad-demo	Rexerai
Hotspot	ExtraElement `hotspot` (vừa upgrade modal)
Slider Compare	ExtraElement `compare`
Auto Rotating Gallery / Swipe Gallery	Block `image_carousel` (Swiper)
Cinemagraph (đơn giản)	ExtraElement `video` với autoplay/muted/loop
Tap to Play	ExtraElement `video` với poster
Shoppable	Block `cta_section` + extra `hotspot`

2.2. 🟢 DỄ MIGRATE — extends existing kind (~1 ngày/cái)

Smart-ad-demo	Đề xuất
Animated Counter	Mở rộng `text` ExtraElement: `content.counterTo`, `counterDuration`, `counterFormat`. Render = count-up animation khi vào view.
Typewriter / Text Scramble / Split Text Reveal	Thêm `text` `content.animation: 'typewriter' | 'scramble' | 'split-slide-up' | 'fade-up'` + `animationDuration`
Marquee Text	Thêm `text` `content.marquee: { speed, direction }`
Tap Reveal Tiles	Mở rộng `photo` thêm `content.revealMode: 'tap-flip'` (CSS rotate-y)
Tap & Hold (video)	Mở rộng `video` `content.holdToPlay: boolean`
Pinch to Zoom (photo)	Mở rộng `photo` `content.zoomable: boolean` (hammer.js or native)
Border Beam (CTA)	Style option cho button: `style.beamColor`, `beamSpeed`
Magnetic Button (CTA)	Style option: `style.magneticStrength`
Glassmorphism	Style option chung: `style.glass: { blur, tint }`

2.3. 🟡 KIND MỚI cần build — independent ExtraElement (~2-3 ngày/cái)

Format	Kind đề xuất	Content shape sơ bộ
Countdown / Live Availability	`countdown`	`{ targetDate, format: 'days-hours-mins-secs' | 'flip', expiredText }`
Scratch Reveal	`reveal` (`mode: 'scratch'`)	`{ coverUrl, revealUrl, brushSize }`
Peel to Reveal	`reveal` (`mode: 'peel'`)	Same shape, different gesture
Draw to Reveal	`reveal` (`mode: 'draw'`)	Same shape
Cursor Reveal (spotlight)	`reveal` (`mode: 'spotlight'`)	`{ baseUrl, revealUrl, spotlightSize }`
Water Ripple	`effect_canvas` (`mode: 'ripple'`)	`{ imageUrl, rippleStrength }` (WebGL shader)
Particle Reveal	`effect_canvas` (`mode: 'particles'`)	`{ imageUrl, particleCount }`
Animated Counter	`counter`	`{ from, to, duration, format, prefix, suffix }` (hoặc gộp vào text variant như §2.2)
Spin the Wheel	`spin_wheel`	`{ segments: [{label, weight, prize}], spinDuration }`
Eyes Follow	`eyes_follow`	`{ pairPositions: [{x,y}], maxRotateDeg }`
Liquid Image	`effect_canvas` (`mode: 'liquid'`)	WebGL displacement on pointer move

2.4. 🟠 BLOCK TYPE MỚI cần build — top-level section (~3-5 ngày/cái)

Format	BlockType đề xuất	Content shape
Stories (Instagram-style)	`stories`	`slides: [{ mediaUrl, mediaType, durationMs, caption }]`
Split Screen	`split_screen`	`leftView: BlockContent, rightView: BlockContent, defaultView: 'left' | 'right'`
Quiz	`quiz`	`questions: [{ prompt, options: [{label, weight}] }], results: [{ scoreRange, title, body, cta }]`
Poll / Survey	`poll`	`question, options: [{label}], showLiveResults`
Tile Selector	`tile_selector`	`tiles: [{ id, label, imageUrl, body }], onSelect: 'expand' | 'navigate'`
Bespoke Selector	`wizard`	Multi-step `steps: [{question, options}]`
Configurator	`configurator`	`dimensions: [{label, options}]`, `previewLayers: [...]` — heavy
Longform Story	`longform_story`	Mixed media `sections: [{mediaUrl, body}]`
Chapter Story	`chapter_story`	`chapters: [{title, body, mediaUrl}]`
3D Carousel	`carousel_3d`	Extends `image_carousel` với CSS transform-style: preserve-3d

Format	BlockType đề xuất	Content shape
3D Flip Gallery	`flip_gallery`	`cards: [{frontUrl, backUrl, frontText, backText}]`
Flipbook	`flipbook`	`pages: [{imageUrl}]` (turn.js-style)
Stories	(đã liệt ở trên)	
Filmstrip	Variant của `image_carousel` (orientation: 'vertical')	1 prop thêm
Pannable (360 panorama)	`panorama`	`{ panoramaUrl, fov, initialYaw }` (PhotoSphere / Three.js)

2.5. 🔴 NẶNG (WebGL / Three.js / specialized) — defer cho phase sau

Format	Lý do defer
AR Camera	getUserMedia + 3D overlay; cần webxr-polyfill
Tilt Explore (gyroscope)	DeviceOrientation API + 3D layers
Interscroller 3D / Scroll Frames	Three.js + 122-frame canvas — đẹp nhưng heavy asset
Hover Frame Sequence	360-spin frame loader + lazy decode
SVG Draw on scroll	SMIL + IntersectionObserver — possible nhưng tốn polish
Deep Parallax (multi-layer)	Drag handles, ScrollTrigger setup phức tạp
Morph Blob	SVG path morph (anime.js / GSAP MorphSVG)
Displacement Gallery	WebGL transition (curtain.js)
Motion Tiles	Many tile micro-animations
Virtual Tour (room walk)	Multi-panorama + navigation graph — gần như sản phẩm riêng
Drag & Drop Planner	Floor-plan editor — gần như sản phẩm riêng
Mini Game (stack blocks)	Game engine micro
Heatmap Explorer	Heatmap shader + interaction layer

2.6. ⛔ KHÔNG khả thi với FE thuần — bỏ qua

Format	Lý do
Voice Control	SpeechRecognition cross-browser kém
Blow to Reveal (mic)	Microphone API, browser permission UX kém
Shake Reveal	DeviceMotion permission iOS yêu cầu HTTPS + prompt
Gravity Physics	DeviceMotion same
Weather Dynamic / Location-Based / Time of Day	BE data source — không thuộc block UI

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3. Recommended migration waves

Wave 1 — Quick visual wins (1 tuần)

Goal: Mở rộng `text` / `photo` đã có, không thêm kind mới.

- Text animation modes: `typewriter`, `scramble`, `split-slide-up`, `fade-up`
- Counter mode trong text (`counterTo`, `counterDuration`)
- Marquee mode trong text
- Photo `tap-to-flip` reveal
- CTA style options: `beam`, `magnetic`, `glass`

Tác động code: tăng ~250 LOC, không đụng BE spec (chỉ thêm trường optional trong `content` / `style` cũ).

Wave 2 — High-impact new kinds (2 tuần)

- `countdown` (Live Availability + flip clock)
- `reveal` (scratch + peel + spotlight, 1 kind 4 modes)
- `counter` (nếu Wave 1 chưa cover)
- `spin_wheel` (gamification, dễ kiếm KPI)

Tác động: types.ts + renderer + toolbar + factory + BE spec update (per-kind §3).

Wave 3 — Block-level lift (3-4 tuần)

- `stories` (Insta) — block type mới
- `split_screen` — block type mới
- `quiz` + `poll` — 2 block types form-like
- `carousel_3d` — extend `image_carousel`

Tác động: BlockType enum mở rộng, factory, renderer, BlockTypeMeta, AddBlockDialog, BE spec.

Wave 4 — WebGL / Three.js (theo nhu cầu)

Chỉ build khi có brief cụ thể từ sale/client. Mỗi cái 1-2 tuần.

Ưu tiên (subjective): 1. `panorama` (real estate use case rõ) 2. `flipbook` (catalog use case rõ) 3. `interscroller_3d` (ad media monetisation) 4. `cinemagraph_pro` (vẫn ext của video)

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4. Mapping sang code locations (cho developer)

Action	File chính
Thêm ExtraElement kind	`block-editor/types.ts` → enum + content interface + ExtraElement union
	`block-editor/extraElements.ts` → DEFAULT_SIZE + factory case
	`block-editor/AddExtraElementButton.tsx` → KINDS array
	`block-editor/ExtraElementOverlay.tsx` → switch case + renderer fn
	`block-editor/ExtraElementToolbar.tsx` → KIND_LABEL/KIND_ICON + editor fn
	`block-editor/ExtraElementsList.tsx` → KIND_META + labelOf case
	`docs/be-extra-elements-spec.md` → §3.X new section
Thêm BlockType	`block-editor/types.ts` → BlockType enum + Content interface + BlockContent union
	`block-editor/factory.ts` → NAME_BY_TYPE + DEFAULTS_VI/EN + createBlock case
	`block-editor/AddBlockDialog.tsx` → ORDERED_TYPES
	`block-editor/LayersPanel.tsx` → ELEMENTS_BY_TYPE entry
	`block-editor/blocks/{NewBlock}.tsx` → React component
	`block-editor/blocks/renderBlock.tsx` → switch case
	`block-editor/BLOCK_TYPE_META` → label + accent color

Action	File chính
	`docs/be-canvas-templates-spec.md` (hoặc tách spec mới)

5. BE delta cần lobby (cho mỗi wave)

Wave	BE work
1	Optional fields trong existing `text` / `photo` schema — additive, không breaking
2	Mỗi kind mới (`countdown` / `reveal` / `counter` / `spin_wheel`) là 1 entry vào ExtraElementKind enum + content schema. BE strict validator phải biết. Cập nhật `docs/be-extra-elements-spec.md` §3 đã có template.
3	BlockType enum mới + content schema cho từng cái. Cập nhật `docs/be-canvas-templates-spec.md` .
4	Có thể cần media pipeline (panorama tile generator, frame extractor).

6. Tracking / analytics (đã có hook)

Mọi element mới nên reuse `ExtraElementTracking` (đã có trong `types.ts`). Public renderer fire `window.dispatchEvent('rexer:track', detail)` — GA/Posthog adapter subscribe.

Quiz / Poll / Spin Wheel đặc biệt cần track sự kiện chi tiết (answer chosen, prize won). Mở rộng `eventName` strategy theo từng kind.

7. Effort estimate tổng

Wave	LOC ước tính	Tuần dev (1 người)
Wave 1	+400	1
Wave 2	+1200	2-3
Wave 3	+2500	4-5
Wave 4	+1500/cái	1-2 mỗi format

Tổng cumulative Wave 1-3: ~4 KLOC + tương đương BE schema work. Sản phẩm đủ cạnh tranh với smart-ad-demo trong 7-9 tuần.

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8. Liên hệ

- FE: @Lantm
- Quyết định thứ tự Wave: cần product owner + sale priority
- BE timeline: phụ thuộc Wave 2+ — BE strict validator hiện chặn schema mới